

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 4
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1. TITLE

Develop a Dynamic Webpage Using JavaScript

2. AIM

To develop a dynamic restaurant webpage using HTML, CSS, and JavaScript that authenticates a customer login, reserves a table, displays chef information, accepts meal pre-orders, and collects customer reviews — all updated dynamically without reloading the page.

3. OBJECTIVE

- To validate a login form using JavaScript before granting page access.
- To dynamically hide and show page sections using `style.display`.
- To read values from text, select, and textarea inputs using `getElementById().value`.
- To update page content dynamically using `innerHTML`.
- To build a table-based restaurant menu using HTML.
- To handle button click events using inline `onclick` handlers.
- To append new content (customer reviews) to a page dynamically.
- To style the interface using an external CSS file, including hover effects.

4. THEORY

JavaScript enables a webpage to respond to user actions and update its content immediately, without requiring a server round-trip or page reload. In this application, JavaScript is used to gate access behind a login form and to power several independent interactive features on a restaurant website.

Conditional Access Control

The `login()` function reads the Name and Email fields using `getElementById().value`. If either field is empty, an `alert()` warns the user and the function returns early. Otherwise, the login box is hidden and the main page is revealed by directly setting each element's `style.display` property to "none" or "block".

Dynamic Content Updates with `innerHTML`

Several functions — `reserveTable()`, `showDish()`, and `orderMeal()` — read a selected value and write a confirmation message into a target element using `innerHTML`. This replaces the element's content

immediately, giving instant feedback without reloading the page.

Appending Content Dynamically

The `addReview()` function is different: instead of replacing content, it uses the `+=` operator on `innerHTML` to append a new `<p>` element containing the review text to the existing `#reviews` container, so every submitted review remains visible alongside earlier ones.

Form Elements and Styling

The page uses text, email, radio, select, and textarea inputs styled through an external stylesheet. CSS rules give inputs, selects, buttons, and the textarea a consistent full-width appearance, and a `:hover` rule darkens the button colour when the pointer is over it.

The basic flow of the application is:

Login Form → Validation (`login()`) → Hide Login / Show Main Page → Table Reservation, Dish Reveal, Meal Ordering, Review Submission → Dynamic DOM Updates (`innerHTML` / `style.display`)

5. ALGORITHM / PROCEDURE

- 1 Create an HTML file with a login box containing Name, Email, a persons dropdown, and an Enter Restaurant button.
- 2 Create a hidden main page container (`style="display:none"`) holding the restaurant content.
- 3 Inside the main page, add a header with the restaurant name and address.
- 4 Add radio buttons for selecting Dine In or Dine Out.
- 5 Add a table-selection dropdown and a Reserve button, plus a message area for the reservation result.
- 6 Add an HTML table listing the restaurant menu with dish names and prices.
- 7 Add a section describing the chef's famous dish, with a button to reveal more detail and a message area.
- 8 Add an unordered list of restaurant staff members.
- 9 Add a meal-selection dropdown, an Order button, and a message area for the order confirmation.
- 10 Add a textarea for customer reviews, a Submit Review button, and a container to display submitted reviews.
- 11 Link an external CSS file to style the header, login box, form controls, buttons, table, and reviews container.
- 12 Create an external JavaScript file and define the `login()` function to validate the Name and Email fields.
- 13 In `login()`, alert the user if a field is empty; otherwise hide the login box and show the main page.
- 14 Define `reserveTable()` to read the selected table and display a confirmation message using `innerHTML`.
- 15 Define `showDish()` to display a description of the chef's famous dish using `innerHTML`.
- 16 Define `orderMeal()` to read the selected meal and display an order confirmation using `innerHTML`.
- 17 Define `addReview()` to validate that the review text is not empty, then append it to the reviews container and clear the textarea.
- 18 Link the JavaScript file to the HTML page using a `<script>` tag.
- 19 Open the page in a browser and test the login form, reservation, dish reveal, meal order, and review submission.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Jagadamba Spice Restaurant</title>
<link rel="stylesheet" href="style.css">
</head>
<body>

<!-- Login Form -->
<div class="login-box" id="loginBox">
<h2>Customer Login</h2>
<input type="text" id="name" placeholder="Enter Name">
<input type="email" id="email" placeholder="Enter Email">
<select id="persons">
<option>1</option>
<option>2</option>
<option>3</option>
<option>4</option>
</select>
<button onclick="login()">Enter Restaurant</button>
</div>

<!-- Main Website -->
<div id="mainPage" style="display:none;">

<header>
<h1>Jagadamba Spice Restaurant</h1>
<h3>Jagadamba Junction, Visakhapatnam</h3>
</header>
<hr>

<h2>Select Dining Option</h2>
<input type="radio" name="type" value="Dine-In"> Dine In
<input type="radio" name="type" value="Takeaway"> Dine Out
<hr>

<h2>Reserve Table</h2>
<select id="table">
<option>Table 1</option>
<option>Table 2</option>
<option>Table 3</option>
<option>Table 4</option>
<option>VIP Table</option>
</select>
<button onclick="reserveTable()">Reserve</button>
<p id="reserveMsg"></p>
<hr>

<h2>Restaurant Menu</h2>
<table>
<tr><th>Dish</th><th>Price</th></tr>
<tr><td>Chicken Biryani</td><td>■280</td></tr>
<tr><td>Mutton Biryani</td><td>■350</td></tr>
<tr><td>Paneer Butter Masala</td><td>■220</td></tr>
<tr><td>Butter Naan</td><td>■40</td></tr>
<tr><td>Fried Rice</td><td>■180</td></tr>
<tr><td>Ice Cream</td><td>■120</td></tr>
</table>
<hr>

<h2>Chef's Famous Dish</h2>
<h3>Special Dum Chicken Biryani</h3>
<p>Prepared by Chef Ravi using authentic spices and slow-cooked dum method.</p>
<button onclick="showDish()">Know Why Famous</button>
<p id="dish"></p>
<hr>

<h2>Restaurant Staff</h2>
<ul>
<li>Head Chef - Ravi Kumar</li>
<li>Assistant Chef - Sai</li>
<li>Restaurant Manager - Ramesh</li>
<li>Captain - Mahesh</li>
</ul>
</div>
</div>
```

```

<li>Service Staff - 12 Members</li>
</ul>
<hr>

<h2>Pre Order Meals</h2>
<select id="meal">
<option>Chicken Biryani</option>
<option>Mutton Biryani</option>
<option>Veg Meals</option>
<option>Paneer Curry</option>
</select>
<button onclick="orderMeal()">Order</button>
<p id="mealMsg"></p>
<hr>

<h2>Customer Reviews</h2>
<textarea id="review" rows="5" cols="50"></textarea>
<br><br>
<button onclick="addReview()">Submit Review</button>
<h3>Reviews</h3>
<div id="reviews"></div>

</div>

<script src="script.js"></script>
</body>
</html>

```

style.css

```

body{
font-family:Arial;
background:#fff8dc;
margin:20px;
}

header{
background:#ff9933;
color:white;
padding:20px;
text-align:center;
}

.login-box{
width:350px;
margin:auto;
margin-top:120px;
padding:20px;
border-radius:10px;
background:white;
box-shadow:0px 0px 10px gray;
}

input,select,button,textarea{
width:100%;
padding:10px;
margin:10px 0;
}

button{
background:green;
color:white;
border:none;
cursor:pointer;
}

button:hover{
background:darkgreen;
}

table{
width:70%;
border-collapse:collapse;
}

table,th,td{
border:1px solid black;
padding:10px;
}

```

```

text-align:center;
}

#reviews{
background:white;
padding:10px;
border-radius:10px;
}

```

script.js

```

function login(){
let name=document.getElementById("name").value;
let email=document.getElementById("email").value;

if(name==" " || email==""){
alert("Please Enter Details");
return;
}

document.getElementById("loginBox").style.display="none";
document.getElementById("mainPage").style.display="block";
alert("Welcome "+name);
}

function reserveTable(){
let table=document.getElementById("table").value;
document.getElementById("reserveMsg").innerHTML=
"Your "+table+" has been reserved successfully.";
}

function showDish(){
document.getElementById("dish").innerHTML=
"Our Dum Chicken Biryani is prepared with premium basmati rice, authentic spices, and slow-cooked
for rich flavor.";
}

function orderMeal(){
let meal=document.getElementById("meal").value;
document.getElementById("mealMsg").innerHTML=
meal+" ordered successfully.";
}

function addReview(){
let review=document.getElementById("review").value;

if(review==""){
alert("Write Review");
return;
}

let div=document.getElementById("reviews");
div.innerHTML+="<p>■ "+review+"</p>";
document.getElementById("review").value="";
}

```

7. CODE EXPLANATION

Code	Explanation
getElementById()	Selects an HTML element using its ID.
.value	Retrieves the current value entered into an input, select, or textarea.
alert()	Displays a pop-up message to the user.
style.display	Shows or hides an element by setting it to "block" or "none".
innerHTML (assignment)	Replaces an element's content with new HTML/text.
innerHTML += ...	Appends new HTML/text to an element's existing content.
onclick="function()"	Runs the given JavaScript function when the button is clicked.

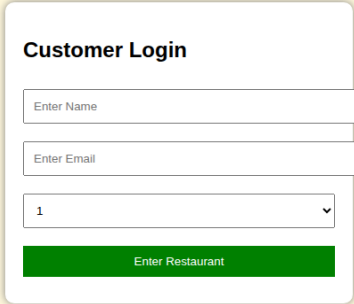
Code	Explanation
if / return	Validates input and exits the function early when a condition is met.
String concatenation (+)	Joins variable values with fixed text to build a message.
:hover (CSS)	Applies styling when the pointer moves over a button.
<table> / <tr> / <th> / <td>	Builds the rows and columns of the restaurant menu.
<select> / <option>	Creates dropdown menus for persons, table, and meal selection.
<textarea>	Provides a multi-line input for writing a customer review.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Click Enter Restaurant with Name and Email empty	Alert should request the user to enter details	Alert displayed correctly	Pass
TC-02	Enter Name "Govindamma", Email, and click Enter Restaurant	Login box hides, main page shows, welcome alert appears	Main page displayed, welcome alert shown	Pass
TC-03	Select "VIP Table" and click Reserve	Confirmation message naming the selected table should appear	"Your VIP Table has been reserved successfully."	Pass
TC-04	Click "Know Why Famous"	Description of the Dum Chicken Biryani should appear	Description displayed correctly	Pass
TC-05	Select "Mutton Biryani" and click Order	Order confirmation naming the selected meal should appear	"Mutton Biryani ordered successfully."	Pass
TC-06	Click Submit Review with the textarea empty	Alert should request the user to write a review	"Write Review" alert displayed	Pass
TC-07	Type a review and click Submit Review	Review should appear in the Reviews list with a star icon	Review appended successfully	Pass
TC-08	Submit a second review	Both reviews should remain visible, one below the other	Both reviews displayed correctly	Pass
TC-09	Hover over any button	Button background should darken	Hover effect applied correctly	Pass

9. OUTPUT

Output 1: Customer Login Screen



A screenshot of a 'Customer Login' form. The form is white with rounded corners and a subtle shadow, centered on a light yellow background. It contains three input fields: 'Enter Name', 'Enter Email', and a dropdown menu with '1' selected. Below these fields is a green button labeled 'Enter Restaurant'.

Screenshot: Login form shown on page load, before Name and Email are entered.

Output 2: Main Page After Login



A screenshot of the main page after a successful login. The page has a light yellow background. At the top is an orange header bar with the text 'Jagadamba Spice Restaurant' and 'Jagadamba Junction, Visakhapatnam'. Below the header is a section titled 'Select Dining Option' with two radio buttons: 'Dine In' and 'Dine Out'. Below this is a section titled 'Reserve Table' with a dropdown menu showing 'Table 1' and a green 'Reserve' button. At the bottom is a section titled 'Restaurant Menu' with a table showing the menu items.

Dish	Price
Chicken Biryani	₹280

Screenshot: Header, dining options, and table reservation section after a successful login.

Output 3: Table Reservation and Restaurant Menu

Select Dining Option

☐

Dine In

☐

Dine Out

Reserve Table

VIP Table

Reserve

Your VIP Table has been reserved successfully.

Restaurant Menu

Dish	Price
Chicken Biryani	₹280
Mutton Biryani	₹350
Paneer Butter Masala	₹220
Butter Naan	₹40
Fried Rice	₹180
Ice Cream	₹120

Screenshot: Reservation confirmation message after selecting "VIP Table" and clicking Reserve.

Output 4: Chef's Famous Dish Reveal

Fried Rice	₹180
Ice Cream	₹120

Chef's Famous Dish

Special Dum Chicken Biryani

Prepared by Chef Ravi using authentic spices and slow-cooked dum method.

Know Why Famous

Our Dum Chicken Biryani is prepared with premium basmati rice, authentic spices, and slow-cooked for rich flavor.

Restaurant Staff

- Head Chef - Ravi Kumar
- Assistant Chef - Sai
- Restaurant Manager - Ramesh
- Captain - Mahesh
- Service Staff - 12 Members

Screenshot: Dish description revealed after clicking "Know Why Famous".

Output 5: Meal Order and Customer Reviews

- Captain - Mahesh
- Service Staff - 12 Members

Pre Order Meals

Mutton Biryani

Order

Mutton Biryani ordered successfully.

Customer Reviews

Submit Review

Reviews

★ Amazing biryani and great service!

Screenshot: Order confirmation for "Mutton Biryani" and a submitted review displayed in the Reviews section.

10. OBSERVATION / INTERVIEW QUESTIONS

Q1. How does the login() function control access to the main page?

Answer: login() reads the Name and Email field values and checks whether either is empty. If validation passes, it sets loginBox's style.display to "none" and mainPage's style.display to "block", so the visual switch between the two sections happens instantly, without navigating to a new page.

Q2. What is the difference between innerHTML assignment and innerHTML += in this program?

Answer: A direct assignment such as document.getElementById("dish").innerHTML = "..." replaces all existing content inside that element. Using += , as in addReview(), appends the new HTML onto whatever content is already there, which is why every submitted review stays visible instead of overwriting the previous one.

Q3. Why does addReview() check if(review=="") before appending?

Answer: This check prevents an empty review from being submitted. If the textarea is empty, the function shows an alert and returns immediately, so nothing is appended to the reviews list.

Q4. What happens to the review textarea after a review is submitted successfully?

Answer: The line document.getElementById("review").value="" clears the textarea back to empty, so the user can immediately type another review without manually deleting the previous text.

Q5. How are the table selection and meal selection values retrieved?

Answer: Both use getElementById(id).value on their respective <select> elements, which returns the text of the currently chosen <option>, such as "VIP Table" or "Mutton Biryani".

Q6. What role does the external CSS file play in this application?

Answer: style.css defines the visual appearance of the page: the orange header, the centred white login card with a shadow, full-width form controls, the green buttons with a darker hover state, and the bordered menu table, keeping the presentation separate from the HTML structure and JavaScript logic.

Q7. Why are the JavaScript functions defined in a separate script.js file instead of inline in the HTML?

Answer: Keeping JavaScript in its own file separates behaviour from structure, making the code easier to read, reuse, and maintain, while the HTML still calls each function through simple onclick attributes on the relevant buttons.

11. RESULT

Thus, a dynamic restaurant webpage was successfully developed using HTML, CSS, and JavaScript. The login form validated user input before granting access, the table reservation, dish reveal, and meal ordering features updated their respective message areas instantly using innerHTML, and the review section successfully appended new customer reviews without clearing previous entries. All features were verified using multiple test cases and the interface was styled consistently using an external stylesheet.